In [108…

```
# Import Libraries
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

In [109…

```
# Load Dataset
df=pd.read_csv("Zomato data .csv")
df.head()
```

Out[109…

	name	online_order	book_table	rate	votes	approx_cost(for two people)	listed_in(type)
0	Jalsa	Yes	Yes	4.1/5	775	800	Buffet
1	Spice Elephant	Yes	No	4.1/5	787	800	Buffet
2	San Churro Cafe	Yes	No	3.8/5	918	800	Buffet
3	Addhuri Udupi Bhojana	No	No	3.7/5	88	300	Buffet
4	Grand Village	No	No	3.8/5	166	600	Buffet

Data Cleaning

In [5]:

```
df.shape
```

Out[5]:

```
(148, 7)
```

In [6]:

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 148 entries, 0 to 147
Data columns (total 7 columns):
 #   Column                       Non-Null Count  Dtype 
---  ------                       --------------  ----- 
 0   name                         148 non-null    object
 1   online_order                 148 non-null    object
 2   book_table                   148 non-null    object
 3   rate                         148 non-null    object
 4   votes                        148 non-null    int64 
 5   approx_cost(for two people)  148 non-null    int64 
 6   listed_in(type)              148 non-null    object
dtypes: int64(2), object(5)
memory usage: 8.2+ KB
```

In [7]:

```
df.duplicated().sum()
```

Out[7]:

```
0
```

In [8]:
```
df.columns = df.columns.str.strip()
```

In [9]:
```
# Remove '/5' and handle missing or invalid values
df['rate'] = df['rate'].astype(str).str.split('/').str[0]

df['rate'] = pd.to_numeric(df['rate'], errors='coerce')
```

In [10]:
```
df.isnull().sum()
```

Out[10]:
```
name                          0
online_order                  0
book_table                    0
rate                          0
votes                         0
approx_cost(for two people)   0
listed_in(type)               0
dtype: int64
```

In [11]:
```
df.dtypes
```

Out[11]:
```
name                            object
online_order                    object
book_table                      object
rate                           float64
votes                            int64
approx_cost(for two people)      int64
listed_in(type)                 object
dtype: object
```

In [12]:
```
df['name']
```

Out[12]:
```
0                       Jalsa
1              Spice Elephant
2             San Churro Cafe
3       Addhuri Udupi Bhojana
4               Grand Village
                ...
143          Melting Melodies
144           New Indraprasta
145              Anna Kuteera
146                    Darbar
147              Vijayalakshmi
Name: name, Length: 148, dtype: object
```

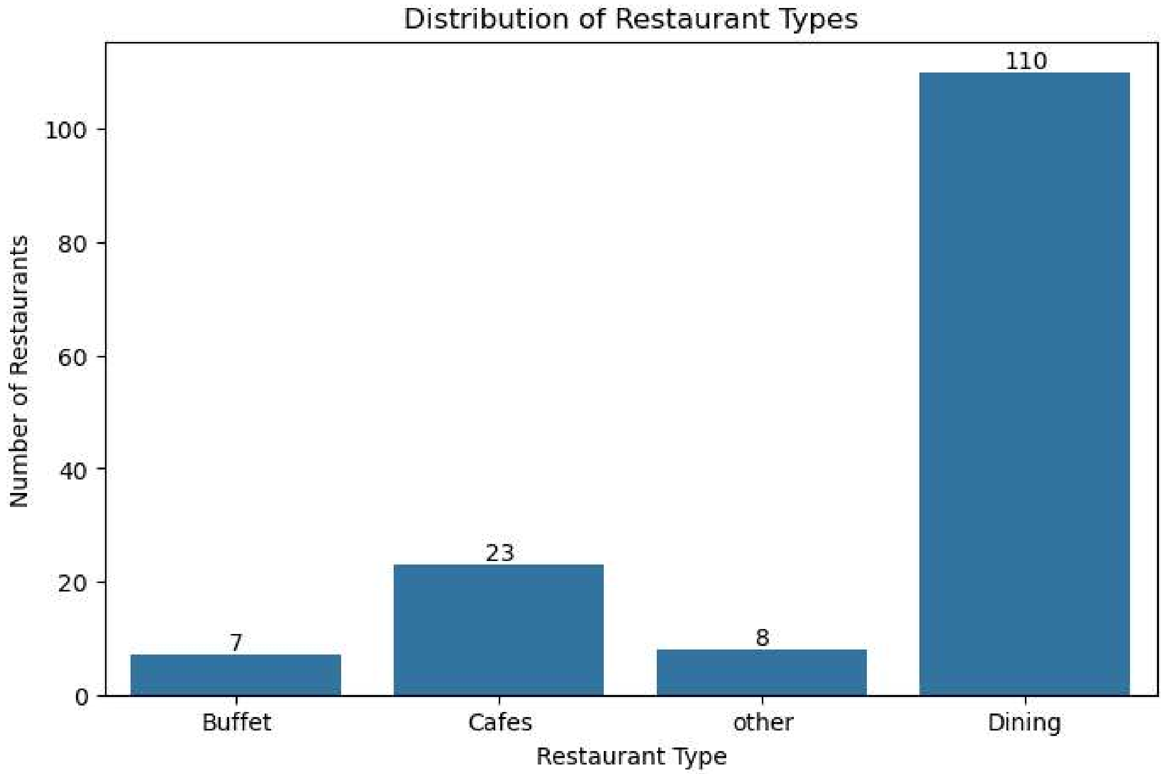
Analysis & Visualization

Analysis 1: Types of Restaurants

1. Use a count plot to show the distribution of restaurant types.

In [119…
```
plt.figure(figsize=(8,5))
ax = sns.countplot(data=df, x='listed_in(type)')
plt.bar_label(ax.containers[0])
plt.title("Distribution of Restaurant Types")
plt.xlabel("Restaurant Type")
```

```
plt.ylabel("Number of Restaurants")
plt.show()
```



Conclusion: Dining is the most common type of restaurant

Analysis 2: Votes by Restaurant Type

In [122...

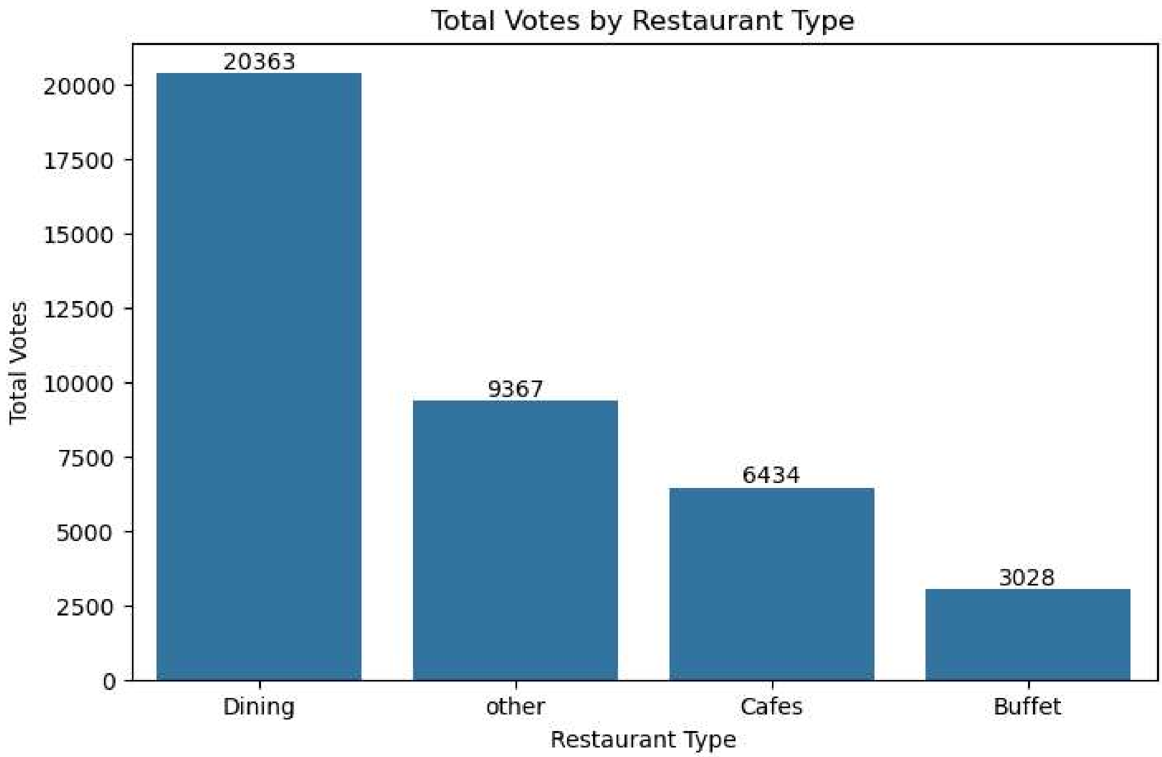
```
a = df.groupby('listed_in(type)')['votes'].sum().sort_values(ascending=False).re
a
```

Out[122...

	listed_in(type)	votes
0	Dining	20363
1	other	9367
2	Cafes	6434
3	Buffet	3028

In [126...

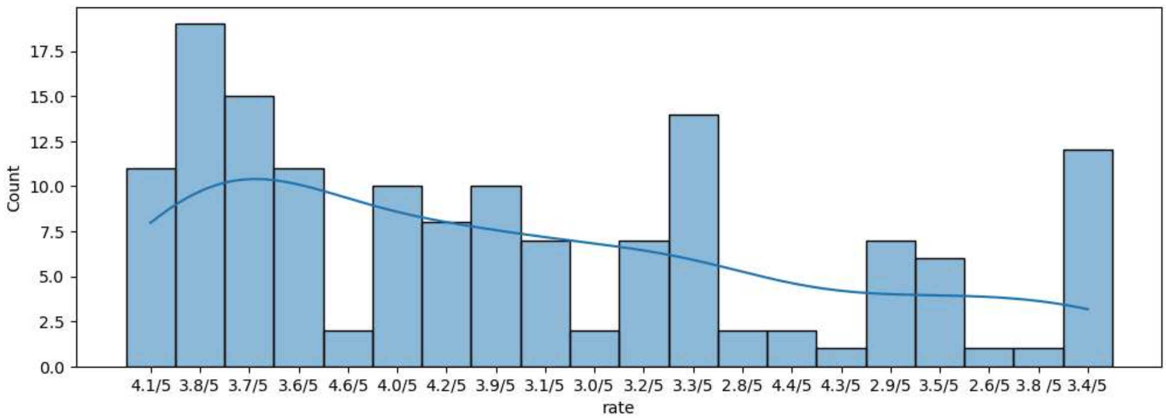
```
plt.figure(figsize=(8,5))
ax = sns.barplot(x='listed_in(type)', y='votes', data=a)
ax.bar_label(ax.containers[0])
plt.title("Total Votes by Restaurant Type")
plt.xlabel("Restaurant Type")
plt.ylabel("Total Votes")
plt.show()
```



Dining restaurants have received the highest total votes

Analysis 3: Ratings Distribution

In [163…
```
plt.figure(figsize=(12,4))
sns.histplot(data=df, x="rate", kde=True, bins = 15)
plt.show()
```

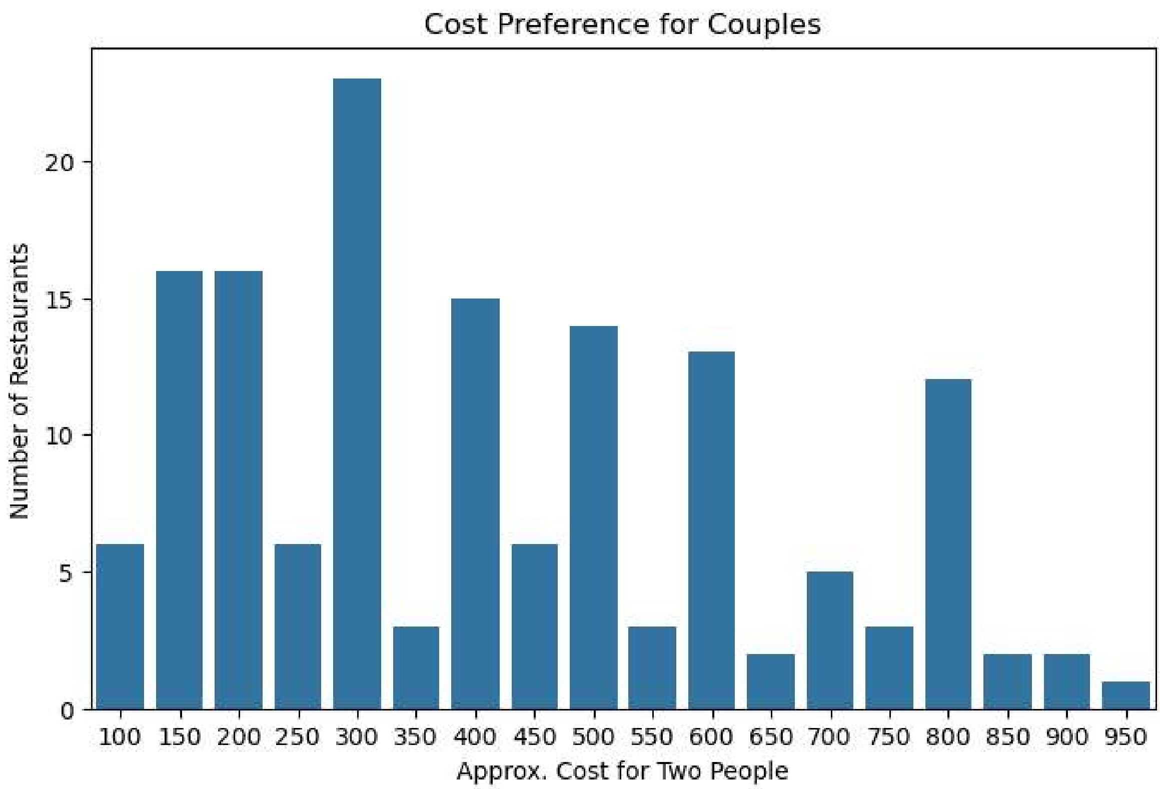


Most restaurants have ratings between 3.8

Analysis 4: Restaurant Cost Preference for Couples

In [167…
```
plt.figure(figsize=(8,5))
plt.title("Cost Preference for Couples")
plt.xlabel("Approx. Cost for Two People")
plt.ylabel("Number of Restaurants")
```

```
sns.countplot(data=df, x='approx_cost(for two people)')
plt.show()
```

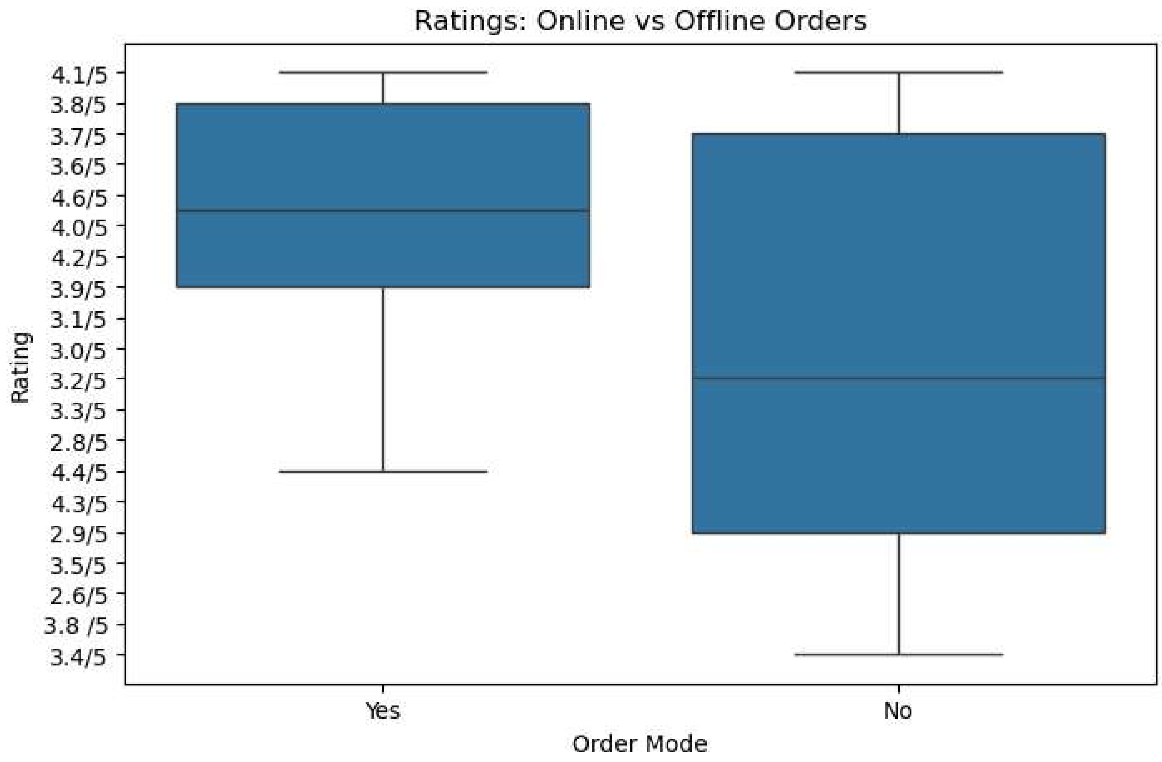


The most common cost for two people is around ₹300-₹800

Analysis 5: Online vs Offline Ratings

In [174…

```
plt.figure(figsize=(8,5))
sns.boxplot(x='online_order',y = 'rate', data=df)
plt.title("Ratings: Online vs Offline Orders")
plt.xlabel("Order Mode")
plt.ylabel("Rating")
plt.show()
```



Cafes have more online orders, while Dining restaurants receive more offline orders.

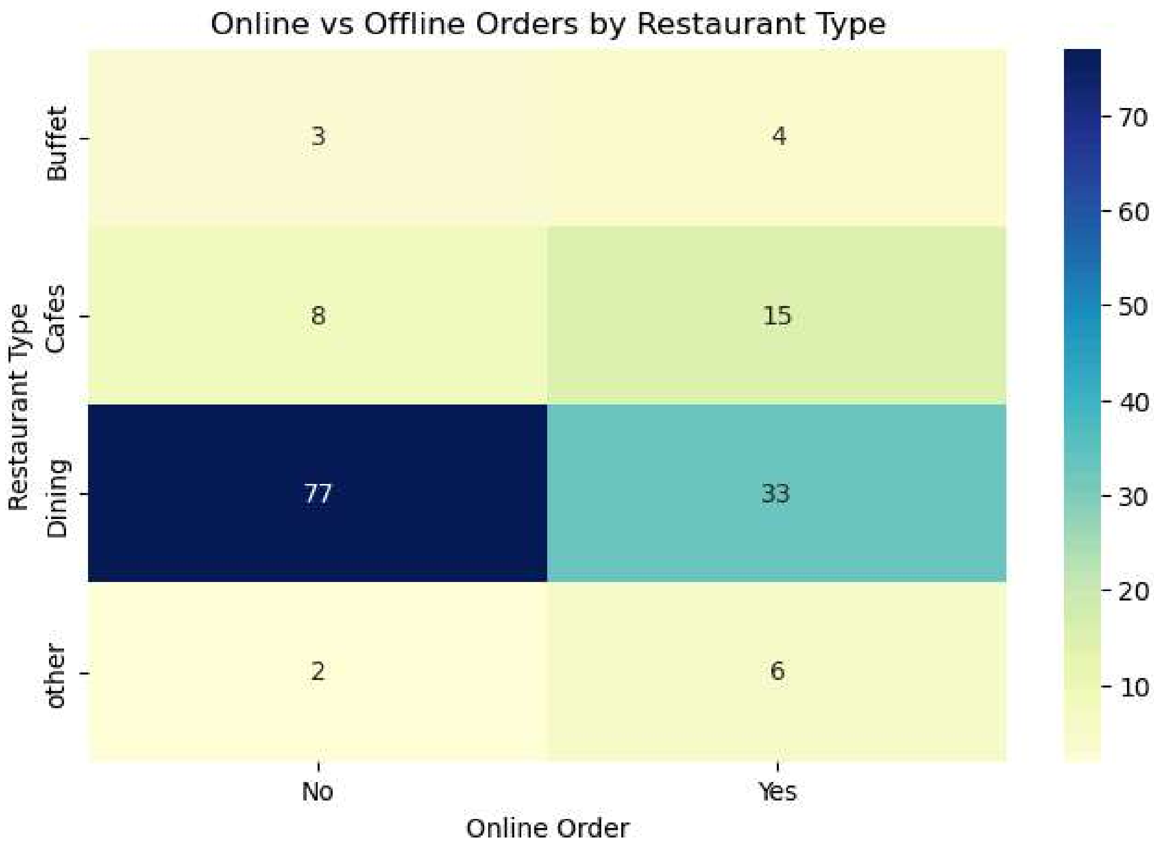
Insights

Online orders have slightly higher ratings compared to offline orders

Analysis 6: Online Orders by Restaurant Type

In [175…

```
pivot_table = pd.crosstab(df['listed_in(type)'], df['online_order'])
plt.figure(figsize=(8,5))
sns.heatmap(pivot_table, annot=True, cmap="YlGnBu", fmt='d')
plt.title("Online vs Offline Orders by Restaurant Type")
plt.xlabel("Online Order")
plt.ylabel("Restaurant Type")
plt.show()
```



Type of restaurant majority order from

What type of restaurant do the majority of customers order from?

In [186…

```
# Find the most common restaurant type
most_common_type = df['listed_in(type)'].mode()[0]
count_common_type = df['listed_in(type)'].value_counts().iloc[0]

print(f"The majority of customers order from '{most_common_type}' type restauran
```

```
The majority of customers order from 'Dining' type restaurants (110 entries).
```

2.) How many votes has each type of restaurant received from customers?

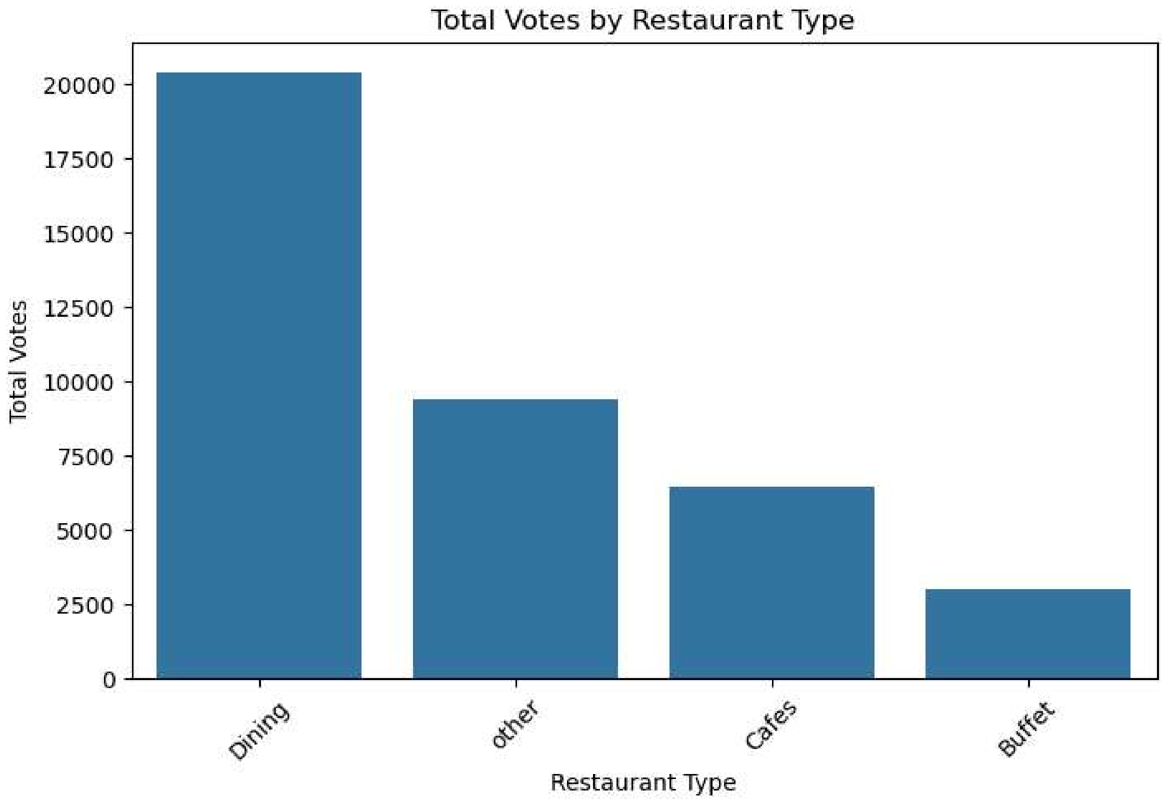
In [188…

```
# Group by restaurant type and sum the votes
type_votes = df.groupby('listed_in(type)')['votes'].sum().sort_values(ascending=

# Display the result
print("Total votes received by each restaurant type:")
print(type_votes)


plt.figure(figsize=(8,5))
sns.barplot(x=type_votes.index, y=type_votes.values)
plt.title("Total Votes by Restaurant Type")
plt.xlabel("Restaurant Type")
plt.ylabel("Total Votes")
plt.xticks(rotation=45)
plt.show()
```

```
Total votes received by each restaurant type:
listed_in(type)
Dining     20363
other       9367
Cafes       6434
Buffet      3028
Name: votes, dtype: int64
```



Majority of votes gets by dining type restaurant

3) Ratings most restaurants received

3.) What are the ratings that the majority of restaurants have received?

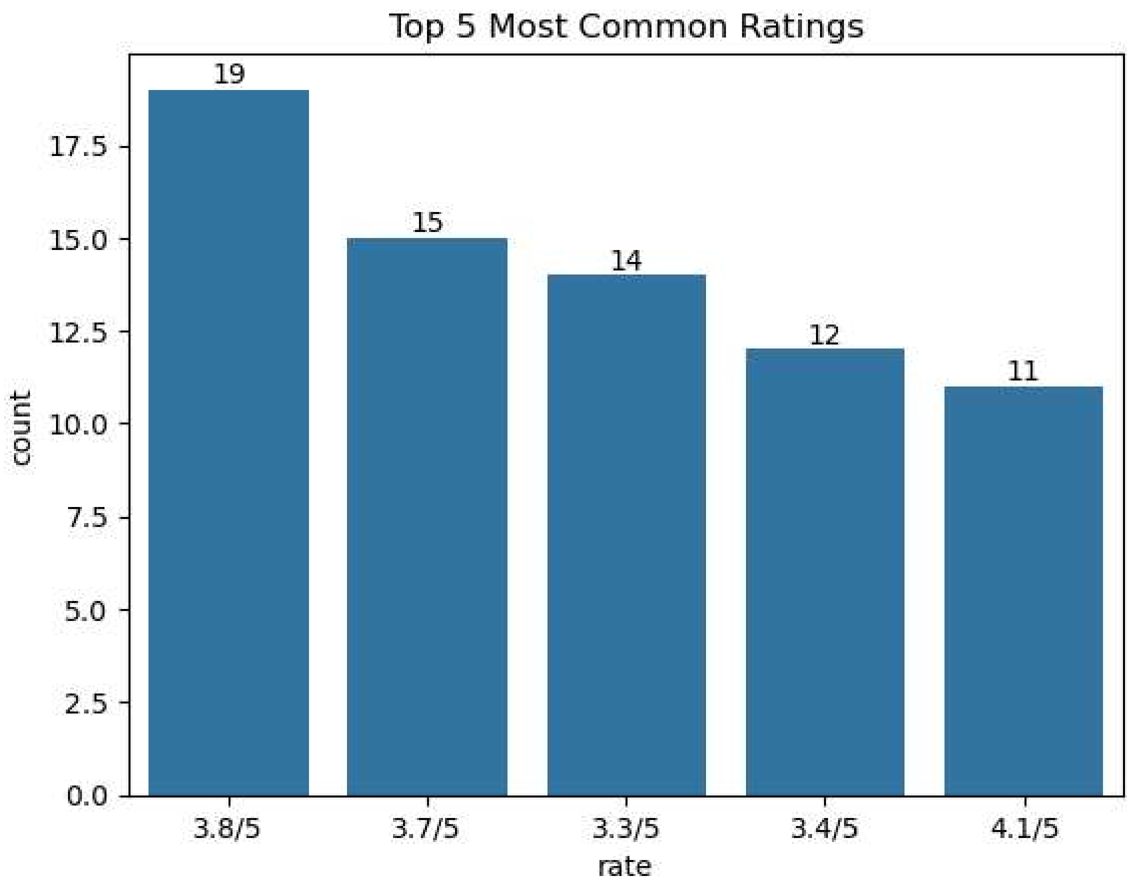
In [189…

```
rating_counts = df['rate'].value_counts().sort_values(ascending=False).head()

# Display top 5 most common ratings
print(rating_counts)


plt.title("Top 5 Most Common Ratings")
ax= sns.barplot(data=rating_counts)
plt.bar_label(ax.containers[0])
plt.show()
```

```
rate
3.8/5    19
3.7/5    15
3.3/5    14
3.4/5    12
4.1/5    11
Name: count, dtype: int64
```



4) Average spending by couples ordering online

4.) Zomato has observed that most couples order most of their food online. What is their average spending on each other?

In [86]:
```
# Filter only restaurants with online orders
online_orders_df = df[df['online_order'] == 'Yes']


# Calculate average spending for two people (couples)
average_spending_two = online_orders_df['approx_cost(for two people)'].mean()

# Average spending per person
average_spending_each = average_spending_two / 2

print("average_spending_two:",average_spending_two, "average_spending_each:",ave
```

```
average_spending_two: 510.3448275862069 average_spending_each: 255.17241379310346
```

5) Mode with maximum rating

5.) Which mode (online or offline) has received the maximum rating?

In [92]:
```
# Group by online_order and find the maximum rating
df.groupby("online_order")["rate"].max()
```

Out[92]:
```
online_order
No     4.3
Yes    4.6
Name: rate, dtype: float64
```

Conclusion: Restaurants with online orders have received the highest maximum rating (4.6).

6) Type with more offline orders

6.) Which type received more offline orders, so that Zomato can provide those customers with some good offers?

In [103…
```
# Filter offline orders
offline_orders_df = df[df['online_order'] == 'No']

# Count offline orders by restaurant type
offline_type_counts = offline_orders_df['listed_in(type)'].value_counts().head()

print(offline_type_counts)
```

```
listed_in(type)
Dining    77
Cafes      8
Buffet     3
other      2
Name: count, dtype: int64
```

In [100…
```
sns.barplot(data=offline_type_counts)
plt.show()
```